

PE-EC701A — Microwave Theory and Technique

Semester VII • 3L:0T:0P • 3 Credits

Module 9 Exam Notes: Microwave Systems & Modern Trends

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

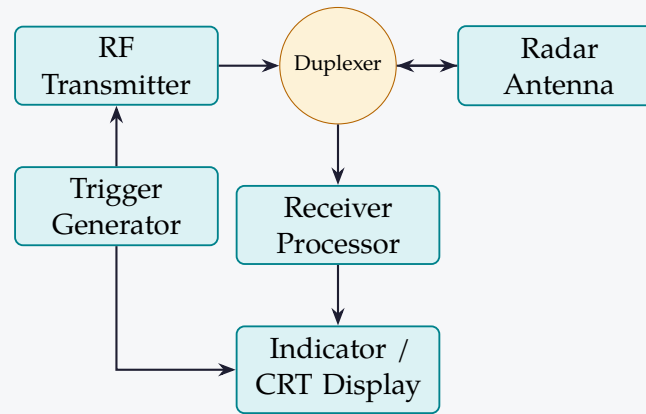
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1. Radar Systems

RADAR (Radio Detection And Ranging) is an active electromagnetic system that transmits microwave pulses and processes reflections to determine target range, velocity, and angle.

Pulsed Radar Block Diagram



Radar Range Equation

The maximum radar detection range $R_{\max}$ is given by:

$$R_{\max} = \left[\frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 P_{\min}} \right]^{1/4} \quad (\text{m}) \quad (1)$$

Where:

P_t = peak transmitter power (W)

G = antenna gain

λ = RF operating wavelength (m)

σ = Radar Cross Section (RCS) of target (m^2)

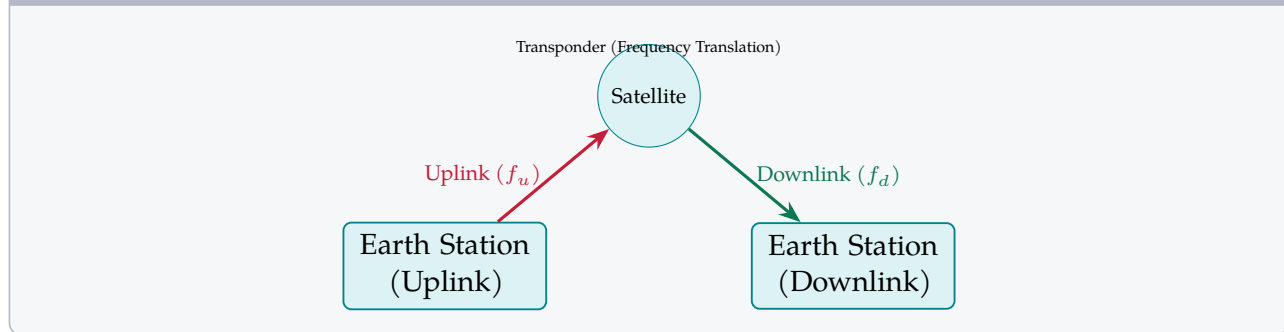
$P_{\min}$ = minimum detectable receiver signal (W)

Radar Variations:

1. **CW (Continuous Wave) Radar:** Transmits continuously, measures velocity using Doppler shift ($f_d = 2v/\lambda$).
2. **FMCW (Frequency Modulated CW) Radar:** Sweeps transmitter frequency, allowing both range and velocity to be measured.
3. **MTI (Moving Target Indicator) Radar:** Suppresses static clutter (ground, buildings) using delay-line cancelers.

2. Terrestrial and Satellite Communications

Satellite Uplink/Downlink Path Schematic



Link Budget (C/N): Carrier-to-Noise ratio at the receiver terminal:

$$\left(\frac{C}{N}\right)_{\text{dB}} = \text{EIRP}_{\text{dBW}} + \left(\frac{G}{T}\right)_{\text{dB/K}} - \text{FSL}_{\text{dB}} - k_{\text{dBW}} - B_{\text{dB-Hz}} \quad (2)$$

where EIRP is Equivalent Isotropic Radiated Power, G/T is receiver figure of merit, $\text{FSL} = (4\pi d/\lambda)^2$ is Free Space Path Loss, and k is Boltzmann's constant.

3. Radio Aids to Navigation

- **GPS (Global Positioning System):** Employs 24+ satellites. Receivers measure transit times of signals from 4+ satellites, using **triangulation** to resolve latitude, longitude, altitude, and clock bias.
- **ILS (Instrument Landing System):** Provides guidance for aircraft landing: **Localizer** (lateral guidance, 108–112 MHz) and **Glide Slope** (vertical guidance, 329–335 MHz).
- **LORAN (Long Range Navigation):** Hyperbolic radio navigation system using time-difference arrivals of pulses from ground stations.
- **RFID (Radio Frequency Identification):** Uses electromagnetic coupling in microwave/RF bands for asset tracking.

4. Modern Trends: Biological Effects, MMICs, and RFMEMS

- **Biological Effects & SAR:** Microwaves cause dielectric heating of tissues. The **SAR (Specific Absorption Rate)** measures RF energy absorption by body tissues:

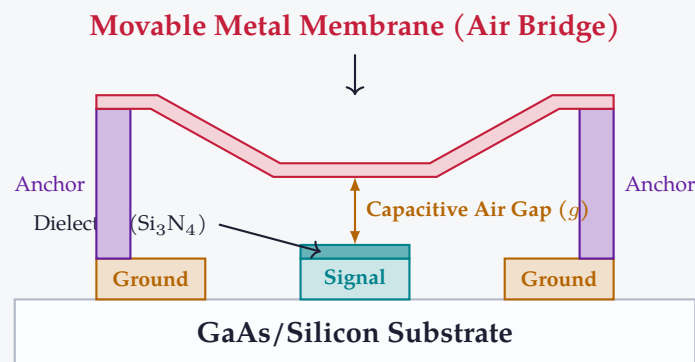
$$\text{SAR} = \frac{\sigma |E|^2}{\rho} \quad (\text{W/kg}) \quad (3)$$

where σ is tissue conductivity, E is electric field, and ρ is mass density. Regulatory limits are typically 1.6 W/kg over 1 g of tissue.

- **MMICs:** Monolithic Microwave Integrated Circuits integrate passive components, transmission lines, and active transistors onto a single semiconductor substrate (GaAs, GaN, SiGe).

- **RFMEMS:** Micro-Electro-Mechanical Systems designed for RF/microwave signals. They offer high isolation, low insertion loss, and low power consumption.
- **Microwave Imaging:** A non-invasive imaging technique that maps the spatial distribution of dielectric properties (permittivity and conductivity) inside a body.
 - **Active Imaging (Radar-based):** Transmits low-power microwave pulses and records the backscattered signals to reconstruct a 3D profile of dielectric contrasts.
 - **Passive Imaging (Radiometry):** Detects natural thermal microwave radiation emitted by the body using ultra-sensitive radiometers.
 - **Applications:** Non-destructive testing, airport security scanners (millimeter-wave body scanning), and medical diagnostics (such as breast cancer screening, which leverages the 5:1 permittivity contrast between high-water-content tumors and normal fat tissue).

RFMEMS Capacitive Switch Model



Cross-sectional view. Electrostatic pull-down voltage collapses the bridge onto the dielectric, switching state from OFF to ON.

Quick Revision — Module IX

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Key Formulas at a Glance

Radar Range Equation	$R_{\max} = \left[\frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 P_{\min}} \right]^{1/4}$
Doppler Frequency Shift	$f_d = \frac{2v}{\lambda} \cos \theta$
Free Space Path Loss	$\text{FSPL} = \left(\frac{4\pi d}{\lambda} \right)^2$
Specific Absorption Rate	$\text{SAR} = \frac{\sigma E ^2}{\rho}$
Shielding Effectiveness	$\text{SE}_{\text{dB}} \approx A_{\text{dB}} + R_{\text{dB}} + M_{\text{dB}}$

One-Line Definitions

- **Duplexer:** A fast waveguide switch permitting a single antenna to be shared by radar transmitter and receiver.
- **Doppler Effect:** The change in frequency of a received wave due to relative motion between source and observer.
- **SAR:** Specific Absorption Rate, measuring the rate of electromagnetic energy absorption per unit mass of human tissue.
- **MMIC:** Monolithic Microwave Integrated Circuit, fabricated entirely on a single semiconductor wafer.
- **RFMEMS:** Micro-Electro-Mechanical Systems utilized for high-performance microwave switching and tuning.
- **Triangulation:** Geometric position mapping utilizing measured transit distances from multiple satellites.
- **Clutter:** Unwanted reflections from ground objects, weather, or buildings that interfere with target radar signals.
- **Microwave Imaging:** A non-invasive imaging method mapping dielectric property contrasts within a target.

Frequently Asked / Viva Questions

Q1. Why is a duplexer critical in pulsed radar systems?

It isolates the sensitive receiver from the high-power transmitter during the pulse transmission, and routes the weak return echo to the receiver during the rest interval.

Q2. How does Doppler shift change with target movement direction?

If the target moves toward the radar, the received frequency increases $(+f_d)$. If it moves away, the received frequency decreases $(-f_d)$.

Frequently Asked / Viva Questions (contd.)**Q3. Explain the term "Specific Absorption Rate" (SAR).**

It is the power absorbed per unit mass of human tissue under RF exposure, measured in W/kg, used to define safety standards for mobile devices.

Q4. What are the advantages of MMICs over hybrid Microwave Integrated Circuits (HMICs)?

Small size, low weight, high reliability, ease of mass production, and elimination of wire bond parasitics.

Q5. Describe the function of localizer and glide path transmitters in ILS.

The localizer provides horizontal guidance relative to the runway centerline. The glide path provides vertical descent guidance to the runway touchdown point.

Q6. Why does Free Space Path Loss (FSPL) increase with frequency?

Because effective antenna aperture decreases as frequency increases ($A_e \propto 1/f^2$), reducing the power intercepted by a matched receiving antenna.

Q7. What are blind speeds in MTI radar and how are they avoided?

Blind speeds are target velocities where the Doppler shift matches the radar pulse repetition frequency, making targets appear static. They are avoided by using staggered PRFs.

Q8. What materials are primarily used to fabricate MMICs and why?

Gallium Arsenide (GaAs) and Gallium Nitride (GaN) due to high electron mobility, which is necessary for gigahertz operation.

Q9. How does a capacitive RFMEMS switch operate?

An electrostatic voltage pulls a suspended metal bridge down onto a dielectric-covered CPW line, dramatically increasing coupling capacitance to short the RF signal.

Q10. Explain how GPS resolves a 3D position using four satellites.

Three satellites locate the receiver at the intersection of three spheres. The fourth satellite resolves the clock synchronization error between the receiver and GPS time.

Q11. State the thermal effects of microwaves on the human body.

Dielectric heating of water molecules in body tissues, which can lead to localized temperature increases and tissue damage (especially in the eyes and testes).

Q12. What is shielding effectiveness (SE) and what are its main mechanisms?

It measures a barrier's ability to attenuate electromagnetic fields, governed by absorption loss, reflection loss, and multiple-reflection corrections.

Q13. Differentiate between active and passive microwave imaging.

Active microwave imaging transmits low-power microwave pulses and measures backscattered signals to reconstruct a dielectric profile. Passive microwave

Frequently Asked / Viva Questions (contd.)

imaging uses highly sensitive radiometers to detect the natural thermal electromagnetic radiation emitted by the body.

Q14. Why is microwave imaging promising for breast cancer detection?

Because malignant tumors have a high water content and thus show a significant dielectric contrast (permittivity and conductivity are about 5 times higher) compared to normal healthy breast fat tissue. This allows clear imaging without ionizing X-ray radiation.

Comparison of MMICs and Hybrid MICs

Parameter	Monolithic Microwave ICs (MMICs)	Hybrid Microwave ICs (HMICs)
Substrate	Single semiconductor wafer (GaAs, GaN).	Alumina, quartz, or Teflon board.
Component Integration	Transistors and matching lines are grown in-situ on the wafer.	Discrete chips are soldered or wire-bonded to substrate tracks.
Reliability	Extremely high (no wire bonds to fail).	Moderate (wire bonds are vulnerable).
Cost (Low Volume)	Very High (complex mask tooling).	Moderate (easy prototyping).
Cost (High Volume)	Very Low (mass batch production).	High (requires manual/auto assembly).